

SHETH L.U.J AND SIR M.V COLLEGE

SUBJECT:- CYBER SECURITY

Practical no: 3

Aim: Implement algorithms to generate and verify message authentication codes (MACs) for ensuring data integrity and authenticity.

Code:

```
import hmac
import hashlib

def generate_mac(message, key):
    return hmac.new(
        key.encode(),
        message.encode(),
        hashlib.sha256
    ).hexdigest()

def verify_mac(message, key, received_mac):
    computed_mac = generate_mac(message, key)
    return hmac.compare_digest(computed_mac, received_mac)

message = input("Enter the message: ")
key = input("Enter the secret key: ")

mac = generate_mac(message, key)

print("\nGenerated MAC:", mac)

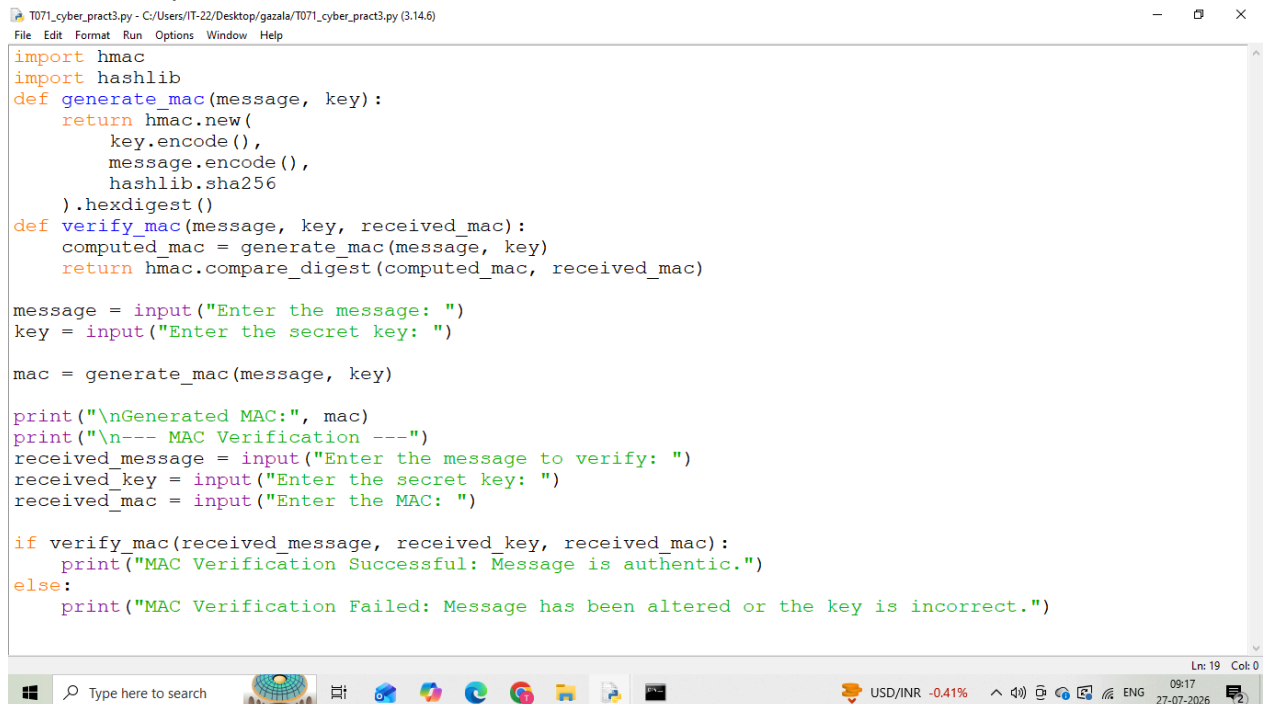
print("\n--- MAC Verification ---")
received_message = input("Enter the message to verify: ")
received_key = input("Enter the secret key: ")
received_mac = input("Enter the MAC: ")

if verify_mac(received_message, received_key, received_mac):
```

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```
print("MAC Verification Successful: Message is authentic.")
else:
    print("MAC Verification Failed: Message has been altered or the key is incorrect.")
```

A screenshot of a text editor window titled 'T071_cyber_pract3.py - C:/Users/IT-22/Desktop/gazala/T071_cyber_pract3.py (3.14.6)'. The editor contains a Python script for MAC verification. The script imports 'hmac' and 'hashlib', defines 'generate_mac' and 'verify_mac' functions, and includes a main execution block with input prompts and conditional printing.

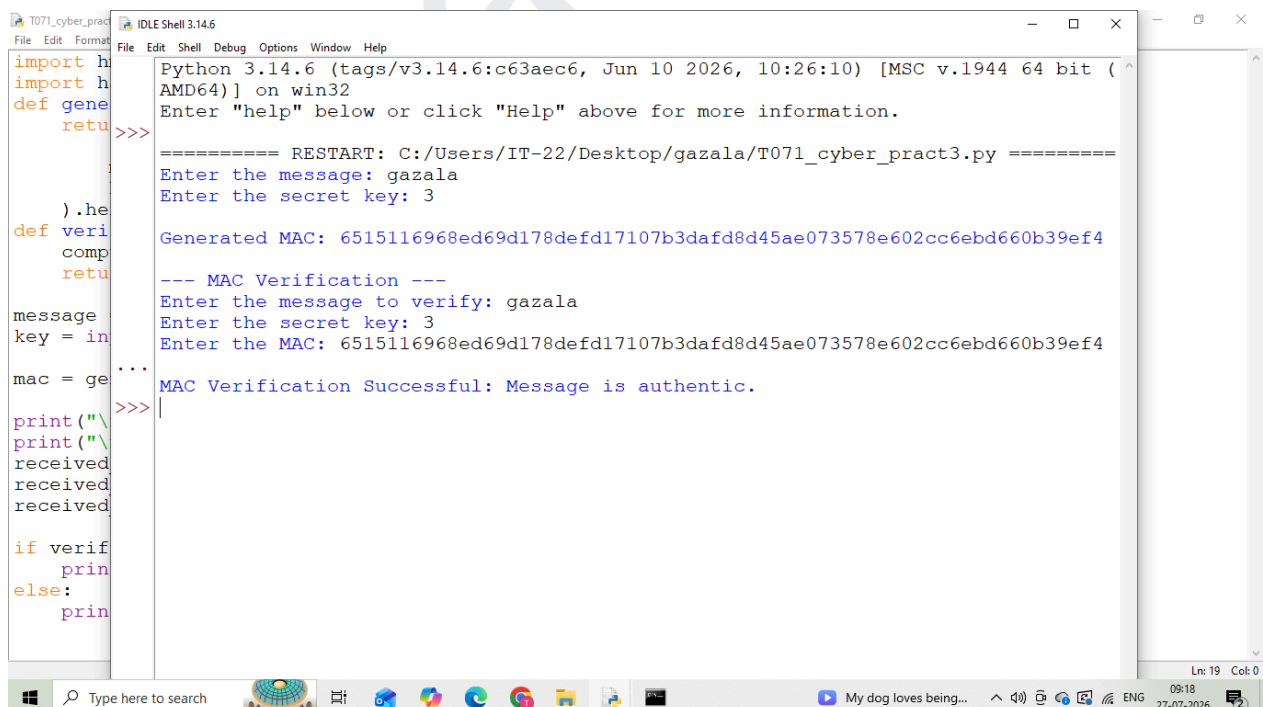
```
import hmac
import hashlib
def generate_mac(message, key):
    return hmac.new(
        key.encode(),
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    ).hexdigest()
def verify_mac(message, key, received_mac):
    computed_mac = generate_mac(message, key)
    return hmac.compare_digest(computed_mac, received_mac)

message = input("Enter the message: ")
key = input("Enter the secret key: ")

mac = generate_mac(message, key)

print("\nGenerated MAC:", mac)
print("\n--- MAC Verification ---")
received_message = input("Enter the message to verify: ")
received_key = input("Enter the secret key: ")
received_mac = input("Enter the MAC: ")

if verify_mac(received_message, received_key, received_mac):
    print("MAC Verification Successful: Message is authentic.")
else:
    print("MAC Verification Failed: Message has been altered or the key is incorrect.")
```

A screenshot showing the execution of the Python script from the previous block in an IDLE Shell 3.14.6. The shell window displays the script's output, including the generated MAC, verification prompts, and the final success message. The taskbar at the bottom shows the system clock as 09:18 on 27-07-2026.

```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

===== RESTART: C:/Users/IT-22/Desktop/gazala/T071_cyber_pract3.py =====
Enter the message: gazala
Enter the secret key: 3

Generated MAC: 6515116968ed69d178defd17107b3dafd8d45ae073578e602cc6ebd660b39ef4

--- MAC Verification ---
Enter the message to verify: gazala
Enter the secret key: 3
Enter the MAC: 6515116968ed69d178defd17107b3dafd8d45ae073578e602cc6ebd660b39ef4

MAC Verification Successful: Message is authentic.
```

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gazala alam